

Choose the best answer and then darken in the answer sheet.

1. Each of the participants _____ ready to play the match.
a) is b) am c) are d) none
2. Don't bite the hands that ☐ ☐ you.
a) feed b) feeds c) have fed d) are feeding
3. The children _____ in the field at present.
a) sit b) bits c) are sitting d) have sat
4. Rosy gave me a bag. Its passive is ☐ ☐.
a) I was given a bag by Rosy. b) A bag was given to me by Rosy.
c) Both a & b d) None
5. If he _____ dinner, he could work properly.
a) has b) had c) had had d) has had
6. Ram Said, "I can pay the money tomorrow." Its indirect speech is ☐ ☐
a) Ram said that he can pay the money tomorrow.
b) Ram said that he could pay the money tomorrow.
c) Ram said that he could pay the money the following day.
d) Ram told to pay the money tomorrow.
7. You always look tired, ☐ ☐ ☐?
a) don't you b) do you c) didn't you d) did you
8. _____ Buddha Air has a good record of flying in the air.
a) A b) An c) The d) None.
9. They never laugh _____ me for my mistakes.
a) in b) on c) by d) at
10. The synonym of the word, "superb" is
a) wonderful b) sunlight c) dangerous d) plight
11. The value of acceleration due to gravity 'g' at the surface of the earth is 10 m/s^2 , what is the value of acceleration due to gravity at height equal to radius of the earth?
a) 1 m/s^2 b) 4.5 m/s^2 c) 10 m/s^2 d) 2.5 m/s^2
12. Which law is given by the statement "If pressure is applied on an enclosed fluid, then it is transmitted equally to all parts of the body"?
a) Archimedes' Principle b) Pascal's law c) Principle of Flotation d) Newton's Law
13. Which of the following is not the effect of gravity?
a) Atmosphere in planet b) Tides in Sea c) Existence of System d) Inertia
14. An object weight 30g in air and 25g when totally immersed in water then the density of the object is
a) 4 g/cm^3 b) 5 g/cm^3 c) 6 g/cm^3 d) 8 gm/cm^3
15. The temperature at which Kelvin scale and Fahrenheit scale coincide is
a) 40°F b) 282.5°F c) 574.1°F d) 642°F
16. What is the average kinetic energy of the molecules contained in a body called?
a) Heat b) Temperature c) Thermal energy d) Internal energy
17. Jack uses the spectacles of power +2D. What is the focal length of the lens?
a) 2m b) 1m c) 0.5m d) 1.5m
18. A step up transformer consisting of 200 turns and 220V of a.c. input is changed into 440V of a.c. output then the no. of turns in the secondary coil is
a) 400 b) 440 c) 220 d) 200

19. Which of the following is scalar quantity?
 a) Current b) Torque c) Angular Momentum d) Acceleration
20. Rihaan can use any source of energy for cooking but he wants to avoid the production of smoke from the source. Which of these sources should he use for cooking?
 a) Coal b) Electricity c) Petroleum d) Wood.
21. The image formed by a concave lens is
 a) Virtual, erect, magnified b) Virtual, erect, diminished
 c) Real, inverted, magnified d) Real, inverted, diminished
22. The force between two bodies having equal masses at a distance is F . If the masses of each two bodies are doubled and distance between their centers is reduced to half then the new force between them is
 a) F b) $2F$ c) $4F$ d) **16F**
23. A body of volume 36cc floats with $\frac{3}{4}$ of its volume submerged in water. The density of body is
 a) 0.25g/cc b) 0.75g/cc c) 0.9g/cc d) 0.1g/cc
24. A piece of ice is floating in a jar containing water. If the temperature of water falls from 25°C to 4°C then the level of water
 a) Rise b) Falls c) Remains unchanged d) None of above
25. How much solar energy is received by every square meter of the earth per second at day time?
 a) 1.6KW b) 1.4KW c) 14KW d) 140KW
26. A body having mass 2kg and specific heat capacity S . If the certain amount of heat is supplied to it, the raise in temperature is 15°C . When the same amount of heat is supplied in 5kg of same substance, the raise in temperature is
 a) 15°C b) 10°C c) 5°C d) 6°C
27. Which physical quantity remains constant during refraction?
 a) wavelength b) velocity c) frequency d) none
28. The thin membrane in the eye through which light enters is known as
 a) Iris b) Pupil c) Cornea d) Aqueous humor
29. Which one is used as heating element in electric appliances?
 a) Chromium b) Nickel c) Nichrome d) Copper
30. A dynamo converts
 a) Electrical energy into mechanical b) Chemical energy into electrical
 c) Mechanical energy into electrical d) Chemical energy into light
31. Silicon shows the characteristics of
 a) Metal b) Non-metal c) Both (a) & (b) d) None
32. Molecular formula of white vitriol is
 a) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ b) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ c) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ d) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
33. Heat is absorbed in _____ reaction.
 a) Endothermic b) Exothermic c) Reversible d) Neutralization
34. The colour of methyl orange in an acid solution is
 a) Colourless b) Pink c) Red d) Yellow
35. Sodium oleate is an example of
 a) Detergent b) Oil c) Pesticide d) Soap
36. What does carbon dioxide form after dissolving in water?
 a) Carbonyl sulphide b) Carbonic acid c) Carbon monoxide d) Carboic acid
37. The alloys of metals with mercury are called
 a) Amalgam b) Bell metal c) Galvanized metal d) Metalloid
38. The general formula of alkyne is.....
 a) $\text{C}_n\text{H}_{2n+2}$ b) C_nH_{2n} c) $\text{C}_n\text{H}_{2n-2}$ d) $\text{C}_n\text{H}_{2n+1}$

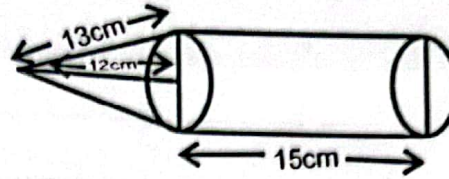
39. Pyrex glass is also called
a) Boro-silicate glass b) Ordinary glass c) Lead crystal glass d) Water glass.
40. Which one of the following increases the setting period of the cement?
a) Gypsum b) Silicon dioxide c) Calcium oxide d) Aluminium oxide
41. Modern periodic table was formulated by
a) Doberener b) Henry Mosely c) Mendeleev d) John Dalton
42. One element M is in the 13th group, the formula of its oxide is
a) MO b) MO₂ c) MO₃ d) M₂O₃
43. The rate of chemical reaction is not affected by
a) Heat b) Light c) Pressure d) Apparatus
44. Which of the following base is used as antacid?
a) Sodium hydroxide b) Potassium hydroxide
c) Calcium hydroxide d) Magnesium hydroxide
45. Sodium carbonate is not used as
a) Ingredients in antacids b) As a cleaning agents
c) For removing hardness of water d) For manufacture of glass
46. The gas prepared by the reaction of ammonium salt with slaked lime is
a) H₂ b) O₂ c) CO₂ d) NH₃
47. The molecular formula of chalcocite is
a) CuFeS₂ b) Ag₂S c) Cu₂S d) Fe₂O₃
48. Which statement is true?
a) CO₂ is basic in nature b) Gold is dissolved in aqua-regia c) PVC is cement d) All of the above
49. Bone meal is
a) Nitrogenous fertilizer b) Phosphorus fertilizer c) Potassium fertilizer d) NPK fertilizer
50. Which one is used in alcoholic beverages for drinking purpose?
a) Methanol b) Ethanol c) Propanol d) All of above
51. What is the study of silkworm called?
a) Melittology b) Sericulture c) Agriculture d) Beekeeping
52. Which of the following fibres carries impulses away from the cyton?
a) Dendron b) Axon c) Cell wall d) Boutons
53. Which part of the brain is responsible for the Posture of our body-
a) Cerebrum b) Cerebellum c) Mid-brain d) Medulla oblongata.
54. Which blood vessels carries deoxygenated blood from the heart to the lungs?
a) Pulmonary artery b) Pulmonary vein c) Venacava d) Aorta
55. Which of the following is normal blood pressure of adults?
a) 120/90 mm of Hg b) 120/70 mm of Hg c) 120/100 mm of Hg. d) 120/80 mm of Hg
56. How many autosome chromosomes are found in human cells?
a) 26 pairs b) 24 pairs c) 23 parts d) 22 pairs
57. Which part of gynoecium forms seed ?
a) Ovary b) Ovule c) Anther d) Pistil
58. In F₂ generation of monohybrid cross, the genotypic ratio is
a) 2:1:1 b) 2:1:2 c) 1:2:1 d) 1:1:2
59. The compound which responsible for ozone layer depletion is
a) Methyl chloroform b) Carbon tetrachloride c) Methyl bromide d) All of above
60. Which of the following plant reproduces by leaf?
a) *Begonia* b) *Bryophyllum* c) Mustard d) Both (a) & (b)
61. Honeybee belongs to phylum
a) Mollusca b) Annelida c) Echniodermata d) Arthropoda
62. The formation of organs or structures in the pupa stage is called

63. The hormone responsible for secondary male sexual characteristics
 a) Hatching b) Spiracles c) Histogenesis d) Histolysis
 a) Estrogen b) Progesterone c) Testosterone d) Glucagon
64. Which of the middle layer of meninges?
 a) Dura mater b) Piamater c) Arachnoid d) Sphenoid
65. The valve present between the right atrium and the right ventricle is the
 a) Tricuspid valve b) Bicuspid valve c) Aortic valve d) Pulmonary valve
66. Excessive number of WBCs in the blood causes
 a) Leukemia b) Anaemia c) Leucopenia d) Tetany
67. Which syndrome is observed due to the lack of one X-chromosome?
 a) Down's syndrome b) Klinefelter's syndrome
 c) Turner's syndrome d) Deficiency syndrome
68. What type of fertilization takes place in frog?
 a) Internal b) External c) Fragmentation d) Triple fusion
69. What is the rooted part of grafting called?
 a) Stock b) Scion c) Stem d) both (a) & (b)
70. What type of variation is mutation?
 a) Continuous b) Discontinuous c) Hereditary d) Environmental
71. If $A \subseteq B$ then $B \cup A$ equals
 a) A b) B c) ϕ d) $B \cap A$
72. If $a = 6 - \sqrt{35}$, find the value of $a + \frac{1}{a}$
 a) -29 b) 1 c) 12 d) 42
73. Area of ΔABC is
 a) $\frac{1}{2} \times \text{base} \times \text{height}$ b) $\frac{1}{2} ab \sin C$ c) $\sqrt{S(S-a)(S-b)(S-c)}$ d) all of the above
74. The inverse of a square matrix A is possible if and only if
 a) $|A| = 0$ b) A is a singular matrix c) A is a non-singular matrix d) $(A) \neq 0$
75. Every rational number is also
 a) a natural number b) a whole number c) an integer d) a real number
76. Find the equation of circle of radius 5 units which touches both the axes and completely lies in the third quadrant.
 a) $x^2 + y^2 - 10x - 10y + 25 = 0$ b) $x^2 + y^2 + 10x + 10y + 25 = 0$
 c) $x^2 + y^2 - 10x - 10y - 25 = 0$ d) $x^2 + y^2 + 10x + 10y - 25 = 0$
77. If $f(x) = x+2$ and $g(x+1) = (x+1)$ then find $\text{gof}(4)$.
 a) 6 b) 10 c) 12 d) 20
78. If $(x+a)$ is a factor of $x^3 + ax^2 - 2x + a + 4$, find the value of a.
 a) $\frac{3}{4}$ b) $\frac{3}{5}$ c) $\frac{5}{3}$ d) $-\frac{4}{3}$
79. The sum of 4th and 8th terms of an AS is 70 and the sum of 6th and 10th term is 94. Find the sum of first 20 terms.
 a) 1200 b) 1204 c) 1240 d) 2140
80. Find the value of k if the lines represented by the equation $(k-9)x^2 - (2k-10)xy + ky^2 = 0$ are coincident to each other.
 a) 25 b) 20 c) 15 d) 10

81. If $\cos\left(\frac{\theta}{2}\right) = \frac{3}{5}$ then the value of $\sin\theta$ is equal to
 a) $\frac{7}{25}$ b) $\frac{10}{25}$ c) $\frac{24}{25}$ d) none
82. If $\vec{p}$ and $\vec{q}$ are unit vectors such that $\vec{p} + 2\vec{q}$ and $5\vec{p} - 4\vec{q}$ are perpendicular to each other, find the angle between $\vec{p}$ and $\vec{q}$.
 a) 30° b) 45° c) 60° d) 90°
83. In a data the quartile deviation and its coefficient are 15 and $\frac{3}{7}$ respectively, find the first quartile.
 a) 10 b) 20 c) 25 d) 30
84. Find A+B when $\tan A + \tan B = 1$ and $\cos A \cdot \cos B = \frac{\sqrt{3}}{2}$
 a) 30 b) 45 c) 60 d) 90
85. If both $px^3 - 3x^2 + 9$ and $x^2 - px + 4$ leaves same remainder when divided by $x+1$, find p.
 a) $-\frac{1}{2}$ b) $\frac{1}{2}$ c) 3 d) -3
86. Test the continuity or discontinuity of the function $f(x) = 2 - x^2$ for $x \leq 2$
 $= x - 4$ for $x > 2$ at $x = 2$ }
 a) continuous b) discontinuous c) limit does not exist d) none
87. The 5th term and 10th term of a geometric sequence are 256 and 8 respectively. Which term is 2?
 a) 5 b) 7 c) 9 d) 12
88. Find the angle between the lines represented by $x^2 - 2xy \cot \alpha - y^2 = 0$
 a) 0° b) 30° c) 60° d) 90°
89. If the inverse of $\begin{pmatrix} x & 1 \\ 5 & y \end{pmatrix}$ is $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$ then find the values of x and y.
 a) 2, 3 b) -2, -3 c) 3, 2 d) none
90. Find the equation of a circle whose centre is (2,7) and which has the radius same as the radius of circle $x^2 + y^2 + 14x + 16y + 24 = 0$
 a) $x^2 + y^2 - 4x - 14y - 36 = 0$ b) $x^2 + y^2 + 4x + 14y + 36 = 0$
 c) $x^2 + y^2 - 4x + 14y + 36 = 0$ d) all of the above
91. The present ages of father and his son are 35 years and 12 years respectively. How many years ago the product of their ages was 210?
 a) 5 b) 7 c) 9 d) 12
92. The value of $\left(1 - \frac{2}{a}\right) \left(1 - \frac{1}{a-2}\right) \left(1 + \frac{3}{a-3}\right)$ is
 a) 0 b) 1 c) 2 d) 6
93. What is the probability of drawing ace or a king from a deck of 52 cards?
 a) $\frac{1}{2}$ b) $\frac{2}{13}$ c) $\frac{5}{13}$ d) $\frac{8}{13}$

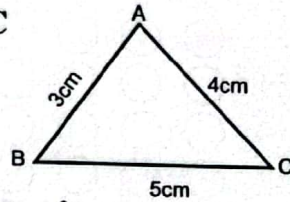
94. The slant height of a cone is 6 times of its base radius. If the volume of the cone is 198cm^3 , find the total surface area of the cone
- a) 122.38cm^2 b) 212.38cm^2 c) 222.38cm^2 d) none

95. Find the volume of the given solid



- a) 1419.5cm^3 b) 1471.5cm^3 c) 1481.5cm^3 d) 1491.5cm^3
96. Ram deposited certain sum of money in a bank at 12% rate of interest p.a. compounded yearly. If he received Rs. 3816 as an interest after 2 years, what was his sum?
- a) Rs. 12000 b) Rs. 14000 c) Rs. 15000 d) Rs. 18000

97. Find the area of $\triangle ABC$



- a) 6cm^2 b) 9cm^2 c) 12cm^2 d) 36cm^2
98. If $\cos\alpha = \frac{3}{5}$ and $\sin\beta = \frac{12}{13}$ then $\sin^2\left(\frac{\alpha - \beta}{2}\right)$ is equal to
- a) $\frac{1}{65}$ b) $\frac{1}{25}$ c) 25 d) 65
99. If $x = 3^{\frac{1}{3}} + 3^{\frac{2}{3}}$, then the value of $x^3 - 9x$ is equal to
- a) 9 b) 10 c) 12 d) 15
100. A tree 100 meters high casts a shadow of 173 meter. Find the altitude of the sun.
- a) 30° b) 45° c) 60° d) 90°

Best of Luck

SHIKSHADEEP B.S.S. +2 Science

ANSWER SHEET

NAME:

SET:

Students are requested to darken the correct answer

Correct Method ☐ ☒ ☐ ☐ Wrong Method ☒ ☒ ☒ ☒

Symbol Number

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